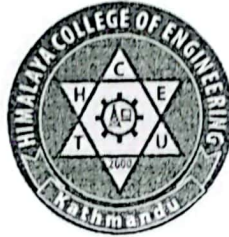
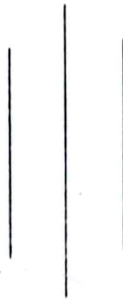




TRIBHUVAN UNIVERSITY
INSTITUTE OF SCIENCE AND TECHNOLOGY



HIMALAYA COLLEGE OF ENGINEERING
CHYASAL, LALITPUR



Lab Report No:- 3

Title:- update, query, Delete query & sub query.

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TITLE : Update Query, Delete Query & sub Query.

OBJECTIVE :

→ The objective of this lab is to perform SQL queries to update, delete and manipulate data within a database using ms SQL. The lab focuses on working with the following tasks: retrieving data using conditions, adding attributes, updating records, performing subqueries and deleting records.

THEORY :

1, Update Query :

→ The UPDATE statement is used to modify the existing records in a table. It allows us to update one or more columns at once based on a specific condition.

Syntax :

UPDATE table_name

SET column1 = value 1, column2 = value2, ---, columnN = valueN

WHERE [condition];

2, Cartesian Join (CROSS JOIN) :

→ The CARTESIAN JOIN or CROSS JOIN produces the cartesian product of two or more joined tables, resulting in every possible combination of rows between the joined tables. It's equivalent to an inner join where the join condition always evaluates to true or when there is no join condition.

Syntax :

SELECT table1.column1, table2.column2, ---

FROM table1, table2 [, table3] ;

1, Delete Query:

The DELETE query is used to remove records from a table. A WHERE clause can be used to delete specific rows; otherwise, all rows from the table will be deleted.

Syntax:

```
DELETE FROM table_name  
WHERE [condition];
```

2, TRUNCATE TABLE: The truncate table command is used to delete all data from an existing table. Unlike the DELETE command, it removes all rows without affecting the table structure. For removing both the data and structure, the DROP TABLE command is used, which requires recreating the table to store new data.

Syntax:

```
TRUNCATE TABLE table_name;
```

SQL subqueries: A subquery is a query within another query. It is typically used in the WHERE clause to return data that is used by the main query to restrict the data retrieved. Subqueries can be utilized with SELECT, INSERT, UPDATE and DELETE statements and can involve various operators like =, <, >, >=, <=, IN, BETWEEN, etc.

Rules for subqueries:

- subqueries must be enclosed within parentheses.
- A subquery can only return one column unless multiple columns are compared in the main query.
- ORDER BY cannot be used in a subquery.

- GROUP BY can be used in place of ORDER BY within a subquery.
- subqueries that return more than one row can only be used with multi-value operators like IN.
- BETWEEN cannot be used with a subquery, but it can be used within a subquery.

subqueries with the SELECT statement : subqueries are most commonly used with the SELECT statement. The main query uses the data returned by the subquery as a condition for filtering records.

syntax :

```
SELECT column_name [, column_name]
FROM table1 [, table2]
WHERE column_name OPERATOR
(SELECT column_name [, column_name]
FROM table1 [, table2]
[WHERE ]);
```

> subqueries with the INSERT statement : subqueries can also be used in conjunction with the INSERT statement to insert data into a table based on the result of another query.

syntax :

```
INSERT INTO table_name [(column1 [, column2])]
SELECT [* | column1 [, column2]
FROM table1 [, table2]
[WHERE VALUE OPERATOR];
```


7, subqueries with the UPDATE statement: A subquery can be used within an UPDATE statement to modify records based on data returned from another query.

Syntax:

```
UPDATE table_name
```

```
SET column_name = new_value
```

```
WHERE [OPERATOR] [VALUE]
```

```
(SELECT column_name FROM TABLE_NAME  
[WHERE]);
```

8, Subqueries with the DELETE statement: Subqueries can be used with DELETE statements to delete records based on the data returned by the subquery.

Syntax:

```
DELETE FROM TABLE_NAME
```

```
WHERE [OPERATOR] [VALUE]
```

```
(SELECT column_name FROM TABLE_NAME  
[WHERE]);
```

9, Correlated subqueries: A correlated subquery is a subquery that refers to columns from the outer query. Unlike a regular subquery, a correlated subquery is evaluated once for every row processed by the outer query.

Syntax:

```
SELECT column_name [, column_name]
```

```
FROM table1
```

```
WHERE column_name OPERATOR
```

```
(SELECT column_name FROM table2
```

WHERE table1.column_name = table2.column_name);

OBSERVATION:

Lab tasks:

1> find all the book names, publication names and author names where the publication name is "Ekta":

→ SQL:

```
SELECT Bookname, PublicationName, AuthorName  
FROM Books  
WHERE PublicationName = 'Ekta';
```

2> find the teacher's name and faculty who issued a book on Jan 1, 2015:

→ SQL:

```
SELECT TeacherName, faculty  
FROM Issues  
WHERE IssueDate = '2015-01-01';
```

3> find the employee name whose salary is greater than 10000 and faculty is "computer":

→ SQL:

```
SELECT EmployeeName  
FROM Employees  
WHERE salary > 10000 AND faculty = 'computer';
```

4> Add an attribute bld to the Issues relation:

→ SQL:

```
ALTER TABLE Issues  
ADD bld INT;
```


5, Insert data into the bid column:

→ SQL:

```
UPDATE Issues  
SET bid = 101  
WHERE IssueID = 1;
```

6, find the teacher's name and book name issued by the teacher whose name starts with 'S':

→ SQL:

```
SELECT TeacherName, BookName  
FROM Issues  
WHERE TeacherName LIKE 'S%';
```

7, update all salaries by 10%:

→ SQL:

```
UPDATE Employees  
SET salary = salary * 1.10;
```

8, update the book name "DBMS" to "DATABASE":

→ SQL:

```
UPDATE Books  
SET Bookname = 'DATABASE'  
WHERE BookName = 'DBMS';
```

9, update the salary of all employees by 20% whose salary is less than 5000:

→ SQL:

```
UPDATE Employees  
SET salary = salary * 1.20  
WHERE salary < 5000;
```

10, Delete the records from the Employee table where empid is 111:

→ SQL:

```
DELETE FROM Employees  
WHERE eid = 111;
```

1, use a subquery to find all teachers names and faculties whose employment date is Jan 2, 2011:

→ SQL:

```
SELECT TeacherName, faculty  
FROM Employees  
WHERE Employee Date = '2011-01-02';
```

12, use a subquery to find all the book names and author names whose publications is "shaja prakashan":

→ SQL:

```
SELECT BookName, AuthorName  
FROM Books  
WHERE PublicationName = (SELECT PublicationName  
FROM publishers WHERE publisherName = 'shaja  
prakashan');
```

DISCUSSION:

→ In this lab, we explored various SQL operations such as cartesian joins, UPDATE queries, DELETE queries and subqueries. The practical tasks covered a wide range of database manipulation techniques including updating records based on specific conditions, deleting records and performing the complex queries using subqueries. By using CROSS JOIN, we understood how a cartesian

product is formed between two tables. The UPDATE and DELETE queries helped us learn how to modify and remove data based on conditions. Additionally, subqueries allowed us to fetch data from one table to influence operations on another, showing the flexibility and power of SQL in handling complex data retrieval & manipulation tasks. The tasks related to salary adjustments and book information updates illustrated real-world applications of these SQL commands. The ability to add new attributes and update records in bulk with conditions is crucial for effective database management.

CONCLUSION:

→ This lab provided a hands-on experience with UPDATE, DELETE and subquery operations in SQL, enhancing our understanding of database manipulation. Through the exercises, we learned how to write efficient queries to handle data retrieval, modification and deletion. These skills are fundamental for managing large datasets in real-world applications, ensuring accurate and efficient data handling.